SCOVILLE SPLICE

SOP-SH-001: Ultra-Hot Pepper Safety, Handling, and Regulatory Compliance

Standard Operating Procedure for the Capsaicin Design Platform Product Line

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SAFETY & COMPLIANCE

1. Purpose and Scope

This Standard Operating Procedure establishes comprehensive safety protocols, handling procedures, consumer warning requirements, labeling standards, and regulatory compliance pathways for the Scoville Splice product line—ten engineered pepper cultivars spanning 3,000,000 to 13,000,000 Scoville Heat Units (SHU). The document applies to all personnel involved in the cultivation, harvest, processing, packaging, distribution, and public demonstration of Scoville Splice products.

The capsaicin concentrations in this product line exceed those of law-enforcement oleoresin capsicum (OC) sprays and approach pharmaceutical-grade purity. The handling requirements described herein reflect this reality: these are not ordinary food products, and conventional food-service safety protocols are insufficient for the upper tiers of the line.

1.1 Applicability

- All Scoville Splice cultivars: Heat 01 through Heat 10
- All forms: fresh fruit, dried whole, dried ground, oleoresin extract, and infused consumer products
- All operational contexts: greenhouse cultivation, field harvest, laboratory analysis, processing facility, packaging, retail, food-service, and public events
- All personnel: employees, contractors, collaborators, and authorized visitors

2. How Hot Is This? — Comparative Pungency Scale

The Scoville Heat Unit (SHU) scale measures the concentration of capsaicinoids—primarily capsaicin (8-methyl-*N*-vanillyl-6-nonenamide) and dihydrocapsaicin—as determined by HPLC analysis per ASTA Method 21.3. One SHU corresponds to approximately 15.0 µg/g total capsaicinoids (ASTA pungency units converted via the Todd–Tominaga correlation).

Table 2-1. Comparative Pungency Reference

PEPPER / PRODUCT	SHU	CAPSAICIN (MG/G)	CONTEXT
Jalapeño	5,000	0.33	Common grocery-store hot pepper
Habanero	350,000	23.3	Considered "very hot" by most consumers
Carolina Reaper	2,200,000	146.7	Former world-record holder (Winthrop University, 2017)
Pepper X	2,693,000	179.5	Current world record (Winthrop University, 2023)
OC Pepper Spray	2,000,000–5,300,000	133–353	Law-enforcement aerosol (oleoresin capsicum)
Heat 01	**3,000,000**	**200**	**1.1x Pepper X — entry-level Scoville Splice**
Heat 05	**6,000,000**	**400**	**2.2x Pepper X — mid-range**
Heat 10	**13,000,000**	**878**	**4.8x Pepper X — 81% pure capsaicin**
Pure Capsaicin	16,000,000	1,000	Theoretical maximum / pharmaceutical standard

Key Observation

Heat 10 operates at 81% of the theoretical capsaicin maximum. At 878 mg capsaicin per gram of dried fruit, the product is closer in pungency profile to a pharmaceutical-grade irritant than to any conventional food pepper. All products Heat 05 and above exceed the pungency of military and law-enforcement OC formulations.

3. Toxicology and Dose-Response

3.1 Capsaicin Pharmacology

Capsaicin ($C_{18}H_{27}NO_3$, MW 305.41 g/mol) is a potent agonist of the transient receptor potential vanilloid 1 (TRPV1) channel. TRPV1 is a nonselective cation channel expressed on C-fiber nociceptors; activation produces intense burning pain, neurogenic inflammation (substance P and CGRP release), and reflexive withdrawal. Capsaicin is lipophilic (log P = 3.04), poorly water-soluble (0.013 g/L at 25°C), and readily absorbed through mucous membranes and compromised skin.

3.2 Lethal Dose Estimates

3.2.1 Animal Data

The acute oral LD_{50} of capsaicin in mice is 47.2 mg/kg body weight (Glinsukon et al., *Toxicol Appl Pharmacol*, 1980). Additional reference values:

Table 3-1. Capsaicin LD_{50} Values by Species and Route

SPECIES	ROUTE	LD_{50} (MG/KG)	SOURCE
Mouse	Oral	47.2	Glinsukon et al., 1980
Mouse	Intraperitoneal	6.5	Glinsukon et al., 1980
Mouse	Intravenous	0.56	Glinsukon et al., 1980
Rat	Oral	161.2	RTECS (Registry of Toxic Effects)
Rat	Dermal	>512	ECHA Registration Dossier
Guinea pig	Oral	120 (estimated)	Monsereenusorn, 1983

3.2.2 Allometric Extrapolation to Humans

Interspecies dose scaling uses the FDA/EPA allometric body-surface-area method. The human equivalent dose (HED) is derived from the animal NOAEL or LD_{50} using the following relationship:

$$\text{HED} = \text{Animal Dose} \times \left(\text{BW}_{\text{animal}} \,/\, \text{BW}_{\text{human}}\right)^{0.25} \qquad \text{(Eq. 1)}$$

Equivalently, using the body-surface-area conversion factor K_m:

$$\text{HED (mg/kg)} = \text{LD}_{50,\text{mouse}} \times \left(K_{m,\text{mouse}} \,/\, K_{m,\text{human}}\right) \qquad \text{(Eq. 2)}$$

$$\text{where } K_{m,\text{mouse}} = 3,\ K_{m,\text{human}} = 37$$

$$\text{HED} = 47.2 \times (3 \,/\, 37) = 3.83 \text{ mg/kg}$$

For a 70 kg human, this yields an estimated lethal dose of:

$$\text{LD}_{50,\text{human}} \approx 3.83 \text{ mg/kg} \times 70 \text{ kg} \approx 268 \text{ mg} \qquad \text{(Eq. 3)}$$

With a 10x interspecies safety factor:

Conservative lethal dose estimate = 268 mg × 10 ≈ **2,680 mg (2.68 g)**

Upper bound (without safety factor):

47.2 mg/kg × 70 kg = **3,304 mg (3.3 g)** pure capsaicin

3.2.3 Product Equivalents at LD_{50}

Table 3-2. Mass of Dried Fruit Required to Reach Estimated LD_{50} (3.3 g capsaicin)

PRODUCT	CAPSAICIN (MG/G)	DRIED FRUIT AT LD_{50} (G)	APPROXIMATE VOLUME
Heat 01	200	16.5	~3 tablespoons ground powder
Heat 03	303	10.9	~2 tablespoons ground powder
Heat 05	400	8.25	~1.5 tablespoons ground powder
Heat 07	547	6.03	~1 tablespoon ground powder
Heat 10	878	3.76	~0.75 tablespoon ground powder

> **Practical Significance**
>
> While voluntary ingestion of 3.76 g of Heat 10 dried fruit (less than a tablespoon) is unlikely in a single sitting due to the extreme pain response, the margin of safety is extraordinarily narrow compared to any other food product. Involuntary exposure (e.g., dust inhalation during grinding, accidental concentrated extract ingestion) poses a genuine acute toxicity risk, particularly via the inhalation and intravenous-equivalent routes.

3.3 Dose-Response Modeling: Hill Equation

The relationship between capsaicin dose and nociceptive response follows the Hill (sigmoidal E_{max}) equation, applicable to both pain-intensity scoring and physiological endpoints (heart rate, blood pressure, lacrimation):

$$E = E_{max} \times D^{n} / (ED_{50}{}^{n} + D^{n}) \quad \text{(Eq. 4)}$$

Where:

- **E** = observed effect (pain score, VAS 0–10; or physiological response magnitude)
- **E_{max}** = maximum achievable effect (asymptote)
- **D** = dose of capsaicin (mg or mg/kg)
- **ED_{50}** = dose producing 50% of E_{max}
- **n** = Hill coefficient (cooperativity index; typically 1.5–2.5 for TRPV1 agonists)

For oral capsaicin in capsule studies (Rollyson et al., 2014), the fitted parameters are approximately: E_{max} = 10 (VAS pain scale), $ED_{50} \approx 5$ mg, $n \approx 2.0$. This steep dose-response curve means that the transition from mild discomfort to maximal pain occurs over a narrow dose range—a critical consideration for consumer safety.

3.4 NOAEL and LOAEL

Table 3-3. No-Observed-Adverse-Effect Level (NOAEL) and Lowest-Observed-Adverse-Effect Level (LOAEL)

ENDPOINT	SPECIES	NOAEL (MG/KG/DAY)	LOAEL (MG/KG/DAY)	STUDY DURATION	SOURCE
GI mucosal irritation	Rat	2.0	10.0	90 days	Kawada et al., 1986
Hepatocellular changes	Rat	5.0	25.0	90 days	Chanda et al., 2008
Body weight reduction	Mouse	1.0	5.0	28 days	Glinsukon et al., 1980
Respiratory irritation	Rat (inhaled)	0.02	0.1	Acute (4 hr)	ECHA Registration
Oral mucosa (human)	Human	0.5	1.5	Single dose	Rollyson et al., 2014

3.5 Acute Effects by Exposure Route

Table 3-4. Acute Capsaicin Exposure: Effects by Route

ROUTE	ONSET	IMMEDIATE EFFECTS	SEVERE/DELAYED EFFECTS	RESOLUTION
Oral	5–30 sec	Severe oropharyngeal burning, profuse salivation, lacrimation, rhinorrhea, diaphoresis	Nausea, emesis, abdominal cramping, diarrhea; rarely: vasovagal syncope, esophageal spasm	30–90 min (pain peak at 5–15 min)
Dermal	1–5 min	Intense localized burning, erythema, edema	Contact dermatitis (Hunan Hand syndrome); secondary transfer to eyes/mucosae via touch	1–4 hr; may persist 12–24 hr without decontamination
Ocular	<10 sec	Immediate blepharospasm, severe pain, profuse lacrimation, conjunctival injection	Corneal epithelial disruption (reversible), chemosis, periorbital edema	30–60 min with irrigation; up to 24 hr without
Inhalation	<30 sec	Cough, choking, laryngospasm, bronchospasm, chest tightness, mucus hypersecretion	Bronchospasm (may be severe in asthmatics), desaturation, respiratory distress; rarely: laryngeal edema	15–45 min; monitor SpO_2 for 2 hr post-exposure

4. Occupational Exposure Limits

4.1 Derivation of the Occupational Exposure Limit (OEL)

No regulatory OEL exists for capsaicin dust in either the OSHA PEL tables or the ACGIH TLV booklet. The following internal OEL is derived from the NOAEL for respiratory irritation using standard uncertainty-factor methodology per EPA/IRIS and WHO IPCS guidelines:

$$\text{OEL} = \text{NOAEL} \;/\; (UF_1 \times UF_2 \times UF_3 \times UF_4) \qquad \text{(Eq. 5)}$$

Where:

- **NOAEL** = 0.02 mg/kg/day (respiratory endpoint, 4-hour acute rat inhalation study)
- $\mathbf{UF_1}$ = 10 (interspecies extrapolation: rat to human)
- $\mathbf{UF_2}$ = 10 (intraspecies variability: sensitive individuals, asthmatics)
- $\mathbf{UF_3}$ = 3 (subchronic to chronic extrapolation; acute study only)
- $\mathbf{UF_4}$ = 1 (NOAEL available; no LOAEL-to-NOAEL conversion needed)

$$\text{OEL} = 0.02 \;/\; (10 \times 10 \times 3 \times 1) = 0.02 \;/\; 300 = 6.67 \times 10^{-5} \text{ mg/kg/day} \qquad \text{(Eq. 6)}$$

For a 70 kg worker inhaling 10 m^3 air per 8-hour shift:

$$\text{Airborne concentration limit} = (6.67 \times 10^{-5} \times 70) \;/\; 10$$

$$= 4.67 \times 10^{-4} \text{ mg/m}^3$$

Internal OEL (8-hour TWA): 0.0005 mg/m^3 (0.5 µg/m^3)

4.2 Short-Term Exposure Limit (STEL)

The 15-minute STEL is set at 5x the 8-hour TWA, consistent with ACGIH excursion-factor methodology for substances with steep dose-response curves:

$$\text{STEL} = \text{OEL} \times 5 = 0.5 \text{ µg/m}^3 \times 5 = \mathbf{2.5 \text{ µg/m}^3} \qquad \text{(Eq. 7)}$$

Table 4-1. Internal Capsaicin Airborne Exposure Limits

METRIC	CONCENTRATION	AVERAGING PERIOD	BASIS
OEL (TWA)	0.5 µg/m³	8 hours	Respiratory NOAEL, UF = 300
STEL	2.5 µg/m³	15 minutes	5x TWA excursion factor
Ceiling	10 µg/m³	Instantaneous	Blepharospasm/bronchospasm threshold

5. Three-Tier Handling System

All Scoville Splice products are classified into three handling tiers based on capsaicin concentration, airborne dust hazard potential, and the severity of foreseeable exposure incidents. Tier assignments determine PPE requirements, facility controls, personnel training, consumer warnings, and documentation.

Table 5-1. Handling Tier Assignments

TIER	PRODUCTS	SHU RANGE	CAPSAICIN (MG/G)	HANDLING CLASSIFICATION
A	Heat 01–04	3,000,000–5,100,000	200–340	Enhanced Food-Grade
B	Heat 05–07	6,000,000–8,200,000	400–547	Chemical Handling
C	Heat 08–10	9,500,000–13,000,000	633–878	Pharmaceutical / Hazmat

5.1 Tier A — Enhanced Food-Grade (Heat 01–04)

PPE Requirements

- Nitrile examination gloves (minimum 5-mil thickness, powder-free)
- Safety glasses with side shields (ANSI Z87.1+)
- Standard food-service apron
- Dedicated footwear (non-porous, closed-toe)

Engineering Controls

- Local exhaust ventilation at grinding/blending stations
- Minimum 6 air changes per hour (ACH) in processing areas
- Separate HVAC zone from general workspace; no recirculation of processing air

Administrative Controls

- Hands washed with vegetable oil then soap before removing gloves
- No face/eye touching while handling product
- Separate storage from non-Scoville Splice ingredients

5.2 Tier B — Chemical Handling (Heat 05–07)

PPE Requirements

- Sealed-fit chemical splash goggles (ANSI Z87.1+ D3/D4 rated)
- NIOSH-approved N95 or higher particulate respirator (fit-tested annually)
- Double-glove: inner 5-mil nitrile + outer 8-mil nitrile or neoprene
- Long-sleeve lab coat or chemical-resistant apron
- Dedicated footwear with shoe covers

Engineering Controls

- Minimum 12 ACH in processing areas
- All grinding and blending operations performed under ventilated enclosure or fume hood
- Negative-pressure processing room relative to adjacent spaces
- Emergency eyewash station within 25 feet (10-second access, ANSI Z358.1)
- Emergency shower within 50 feet

Administrative Controls

- **Buddy system:** minimum two trained personnel present during all handling operations
- Pre-task safety briefing documented for each processing session
- Maximum continuous handling time: 2 hours, followed by 30-minute break
- Incident response plan posted at each workstation

5.3 Tier C — Pharmaceutical / Hazmat (Heat 08–10)

> **Maximum Hazard Classification**
>
> Tier C products contain capsaicin concentrations between 63% and 81% of the pharmaceutical pure standard. Exposure incidents at this tier can produce effects indistinguishable from chemical weapons exposure. Full hazmat-grade protocols apply.

PPE Requirements

- Full-face NIOSH-approved P100 respirator (fit-tested annually; quantitative fit factor ≥500)
- Tyvek or equivalent chemical-resistant full-body coverall
- Double-glove: inner 5-mil nitrile + outer 15-mil butyl rubber or neoprene (extended-cuff)
- Chemical-resistant boot covers, sealed at ankles
- All seams sealed with chemical-resistant tape at wrist, ankle, and collar interfaces

Engineering Controls

- Minimum 20 ACH in processing areas
- All operations performed inside Class II biological safety cabinet or equivalent containment enclosure
- Negative-pressure processing room with HEPA-filtered exhaust
- Continuous airborne particulate monitoring (real-time OPC or gravimetric sampler)
- Emergency shower and plumbed eyewash station within 10 feet of each workstation
- Spill containment kit (absorbent, neutralizing agent, dedicated disposal containers)

Administrative Controls

- **Buddy system mandatory**—minimum two trained and PPE-equipped personnel
- Pre-task safety briefing with signed acknowledgment
- Maximum continuous handling time: 1 hour, followed by mandatory 30-minute decontamination break
- All processing sessions logged with start/end time, personnel, and environmental monitoring readings
- Product stored in locked, access-controlled enclosure when not in active use

6. Facility Requirements

6.1 PPE Material Performance

Capsaicin, as a lipophilic small molecule, permeates many common glove materials. The following breakthrough-time data guide material selection for extended contact:

Table 6-1. Glove Material Permeation Resistance to Capsaicin Solution (1% w/v in ethanol)

MATERIAL	THICKNESS (MIL)	BREAKTHROUGH TIME (MIN)	PERMEATION RATE (MG/CM²/MIN)	SUITABILITY
Latex (natural rubber)	5	8–12	12.4	Not recommended
Nitrile (standard)	5	25–40	3.2	Tier A (change every 30 min)
Nitrile (heavy-duty)	8	55–80	1.1	Tier B outer glove
Neoprene	12	90–120	0.4	Tier B/C outer glove
Butyl rubber	15	>240	<0.05	Tier C outer glove (preferred)
Viton (fluoroelastomer)	12	>480	<0.01	Extended operations, extract handling

6.2 Ventilation Requirements

Table 6-2. Minimum Air Changes Per Hour (ACH) by Tier and Activity

ACTIVITY	TIER A	TIER B	TIER C
Storage (ambient, sealed containers)	4	6	10
Weighing / portioning	6	12	20
Grinding / blending (dust-generating)	8	15	25
Extraction (solvent-based)	N/A	20	30
Greenhouse cultivation	4	6	8
Harvest (fresh fruit)	Outdoor/natural	4 (enclosed)	6 (enclosed)

6.3 Storage Requirements

- **Tier A:** Labeled, sealed containers in a dedicated storage area; separated from non-Scoville Splice ingredients; ambient temperature (<25°C), low humidity (<60% RH)
- **Tier B:** Secondary containment (sealed bag within sealed container); restricted-access storage room with ventilation; SDS posted

- **Tier C:** Locked, access-controlled cabinet or enclosure; inventory log maintained with chain-of-custody documentation; temperature-monitored (<20°C); SDS posted; spill kit immediately adjacent

7. Consumer Warning Labels

The following warning texts are required on all consumer-facing packaging. Text must appear in a minimum 10-point sans-serif font on a contrasting background. Warning labels must not be obscured by packaging seams, shrink wrap, or adhesive closures.

7.1 Tier A Warning Label (Heat 01–04)

WARNING: EXTREME HEAT

This product contains capsaicin concentrations exceeding 200 mg/g—more than 600 times hotter than a jalapeño and significantly hotter than the Carolina Reaper. Use in very small quantities. Avoid contact with eyes, open wounds, and sensitive skin. Wash hands thoroughly after handling. Not recommended for individuals with gastrointestinal disorders, cardiovascular conditions, or capsaicin sensitivity. Keep out of reach of children. **For adults (18+) only.**

7.2 Tier B Warning Label (Heat 05–07)

DANGER: EXTREME PUNGENCY

This product contains capsaicin at concentrations exceeding 400 mg/g—comparable to or exceeding law-enforcement oleoresin capsicum (pepper spray) formulations. Handle with chemical- rated gloves and eye protection. Do not inhale dust or powder. Ingestion of undiluted product may cause severe pain, nausea, vomiting, and respiratory distress. Use only as directed and in extremely small quantities. Not intended for direct consumption without dilution. Keep out of reach of children and unauthorized persons. **For adults (18+) only.**

Poison Control: 1-800-222-1222 (US) | **Emergency:** 911

7.3 Tier C Warning Label (Heat 08–10)

DANGER: PHARMACEUTICAL-GRADE CAPSAICIN CONCENTRATION

This product contains capsaicin at concentrations between 633 and 878 mg/g (63–81% pure capsaicin). These concentrations exceed all commercially available pepper products and law- enforcement formulations. This product must be stored under lock and handled exclusively by trained adults wearing appropriate protective equipment including respiratory protection, sealed eye protection, and chemical-resistant gloves.

Exposure may cause: severe chemical burns to mucous membranes, bronchospasm, laryngospasm, and incapacitating pain. Direct ocular exposure may cause temporary blindness and corneal injury. Inhalation of dust may cause respiratory distress requiring medical intervention.

Do not open, handle, or use this product without reading the full Safety Data Sheet.

Poison Control: 1-800-222-1222 (US) | **Emergency:** 911

Store locked. For trained adults (18+) only. Unauthorized access prohibited.

8. Regulatory Pathway

8.1 United States

8.1.1 FDA — 21 CFR

Capsaicin derived from *Capsicum* species is Generally Recognized As Safe (GRAS) for use as a flavoring agent (21 CFR 182.10, 182.20). The FDA does not impose an upper concentration limit for capsaicin in food products. However, FDA enforcement discretion under 21 CFR 110 (Current Good Manufacturing Practice) and the FSMA Preventive Controls Rule (21 CFR 117) requires that foreseeable hazards—including acute toxicity from high capsaicin concentrations—be identified and controlled in the food safety plan. Products Heat 05 and above should include capsaicin concentration as a chemical hazard in the HACCP/preventive-controls analysis.

8.1.2 USDA APHIS — SECURE Rule (7 CFR 340, effective 2020)

The USDA Animal and Plant Health Inspection Service regulates genetically engineered organisms under the SECURE Rule (Sustainable, Ecological, Consistent, Uniform, Responsible, Efficient). The regulatory status depends on the type of genetic modification:

- **Products 01–04 (gene deletion / loss-of-function edits only):** These cultivars use targeted knockouts of pungency-suppression pathways via SDN-1 (site-directed nuclease type 1) edits that

introduce no foreign DNA. Under the SECURE Rule's exemption criteria (7 CFR 340.1(b)), single-base deletions and small insertions/deletions that could have been achieved through conventional breeding are **likely exempt** from APHIS notification requirements. A Regulatory Status Review (RSR) request (Am I Regulated?) should be submitted for confirmation.

- **Products 05–10 (transgenic insertions / heterologous gene expression):** These cultivars incorporate genetic material from non-*Capsicum* donor species to enhance capsaicinoid biosynthesis beyond the natural range. These modifications **require APHIS notification** under the permit or notification pathway (7 CFR 340.4). A Permit Application or Notification must be submitted at least 30 days (notification) or 120 days (permit) prior to environmental release.

8.1.3 NBFDS — Bioengineered (BE) Food Disclosure

The National Bioengineered Food Disclosure Standard (7 CFR 66), administered by the USDA Agricultural Marketing Service (AMS), requires disclosure for foods that contain detectable modified genetic material. All Scoville Splice products contain detectable modified DNA and must carry one of the approved disclosure formats: text, symbol, electronic/digital link, or text message.

8.1.4 EPA

The EPA does not regulate capsaicin in food products. Capsaicin-based pesticide products are registered under FIFRA, but Scoville Splice products are food items, not pesticides, and do not fall under EPA jurisdiction.

8.2 Regulatory Summary Table

Table 8-1. US Regulatory Status by Product Tier

TIER	PRODUCTS	EDIT TYPE	APHIS STATUS	FDA	BE LABEL REQUIRED
A	Heat 01–04	SDN-1 (deletion only)	Likely exempt (submit RSR)	GRAS; HACCP plan required	Yes (detectable modified DNA)
B	Heat 05–07	Transgenic insertion	Notification required (30+ days)	GRAS; HACCP plan required; enhanced warnings	Yes
C	Heat 08–10	Transgenic insertion	Permit required (120+ days)	GRAS; HACCP plan required; pharmaceutical-grade warnings	Yes

8.3 International Regulatory Landscape

Table 8-2. International Regulatory Status for Gene-Edited Capsicum Products

JURISDICTION	GOVERNING FRAMEWORK	SDN-1 PRODUCTS (01–04)	TRANSGENIC PRODUCTS (05–10)
European Union	Directive 2001/18/EC; ECJ ruling C-528/16 (2018)	Regulated as GMO (ECJ ruled mutagenesis techniques are not exempt); requires Part C authorization	Full GMO authorization required (Part C deliberate release + Regulation (EC) 1829/2003 for food/feed)
Canada	Novel Food Regulations (Food and Drugs Act, Division 28)	Likely classified as Novel Food; pre-market notification required to Health Canada	Novel Food notification required; environmental assessment under CEPA for plants with novel traits (PNTs)
Japan	MAFF/MHLW notification system (2019)	SDN-1 edits (no template DNA): exempt from GMO regulation; notification to MHLW for food safety	Full Cartagena Act approval required for environmental release and food/feed safety assessment
Australia	Gene Technology Act 2000; OGTR amendment (2019)	SDN-1 edits classified as non-GMO (OGTR 2019 amendment); no license required	Regulated as GMO; OGTR license required for dealings (DIR application)
Brazil	CTNBio Resolution 16/2018	Case-by-case assessment; SDN-1 without template DNA typically classified as non-GMO	Full CTNBio commercial approval required
Argentina	CONABIA Resolution 173/2015	Process-based assessment; SDN-1 products typically classified as non-GMO if no novel combination of genetic material	Full GMO authorization required under Resolution 701/2011

9. First Aid and Decontamination Protocols

> **Critical Principle**
>
> Capsaicin is lipophilic. Water alone does not dissolve it and may spread the contamination. All decontamination protocols begin with a lipophilic solvent (oil, dairy fat, or alcohol) before aqueous washing, except for ocular exposure, where immediate copious water irrigation takes priority to prevent corneal damage.

9.1 Oral Exposure

1. Administer whole milk, cream, yogurt, or ice cream immediately. The casein protein in dairy products binds capsaicin via hydrophobic interaction and strips it from mucosal TRPV1

receptors.

2. If dairy is unavailable, administer vegetable oil (swish and spit) or a sugar solution (10–20% sucrose).
3. **Do not induce vomiting**—re-exposure of the esophageal and oropharyngeal mucosa to capsaicin on emesis worsens injury.
4. Monitor for signs of vasovagal syncope (lightheadedness, bradycardia, diaphoresis). Position supine with legs elevated if symptoms occur.
5. If breathing difficulty develops or if more than 500 mg capsaicin-equivalent was ingested, call 911 and transport to emergency department.

9.2 Dermal Exposure

1. Apply vegetable oil or mineral oil liberally to the affected area. Massage gently for 30–60 seconds to dissolve capsaicin from the stratum corneum.
2. Wipe oil away with dry paper towels or cloth.
3. Wash with dish soap (high-surfactant formulation) and lukewarm water. Repeat 3x.
4. **Do not apply water first**—water without prior oil treatment spreads capsaicinoids across unaffected skin.
5. If burning persists after decontamination, apply a paste of baking soda and water, or a topical anesthetic (lidocaine 2–4%).
6. For large-area exposure (>10% body surface area) or blistering: treat as chemical burn, irrigate with emergency shower, and seek medical attention.

9.3 Ocular Exposure

1. **Begin copious water irrigation immediately.** This is the one exception to the oil-first principle—delay increases corneal damage risk.
2. Remove contact lenses if present (they trap capsaicin against the cornea).
3. Irrigate continuously with clean water or saline for a minimum of 15 minutes. Use an eyewash station (ANSI Z358.1 compliant) if available.
4. Hold eyelids open during irrigation to ensure flushing of the conjunctival fornices.
5. After irrigation, apply a bland ophthalmic solution if available.
6. Seek ophthalmologic evaluation for all Tier B and Tier C ocular exposures, and for any exposure with persistent pain, photophobia, or visual changes after 30 minutes.

9.4 Inhalation Exposure

1. Move the affected person to fresh air immediately. Remove from the contaminated area.
2. If the person is coughing but breathing: sit upright, administer supplemental oxygen if available, and monitor.
3. If bronchospasm develops (wheezing, chest tightness, difficulty breathing): administer albuterol inhaler (2 puffs) if available and the person can coordinate inhalation.
4. If laryngospasm or severe respiratory distress occurs: call 911. Prepare for advanced airway management.
5. Monitor pulse oximetry (SpO_2) for a minimum of 2 hours after any symptomatic inhalation exposure.
6. All Tier C inhalation exposures require emergency medical evaluation regardless of symptom severity.

9.5 Decontamination Agents

Table 9-1. Decontamination Agents and Mechanisms of Action

AGENT	ACTIVE COMPONENT	MECHANISM	EFFECTIVENESS	APPLICABLE ROUTES
Whole milk / cream	Casein (phosphoprotein)	Hydrophobic binding: casein micelles encapsulate capsaicin via nonpolar interactions, stripping it from TRPV1 receptors	High (oral); moderate (dermal)	Oral, dermal
Vegetable oil	Triglycerides	Like-dissolves-like solubilization; capsaicin partitions into the oil phase (log P = 3.04)	High (dermal); moderate (oral swish)	Dermal, oral
Ethanol (rubbing alcohol, spirits)	Ethanol (40–70%)	Co-solvent solubilization; ethanol disrupts capsaicin–receptor binding and increases aqueous solubility	Moderate; may increase absorption if skin is compromised	Dermal (intact skin only)
Mineral oil	Liquid paraffin	Dissolution of capsaicin into nonpolar hydrocarbon phase	High (dermal)	Dermal
Dish soap (surfactant)	Sodium lauryl sulfate / related anionic surfactants	Micelle formation around capsaicin molecules, enabling aqueous rinse removal	Moderate to high (after oil pre-treatment)	Dermal (follow-up wash)
Sugar solution (10–20%)	Sucrose	Competitive TRPV1 receptor interaction (partial); osmotic dilution effect	Low to moderate (inferior to dairy)	Oral (when dairy unavailable)
Water (copious)	H_2O	Physical dilution and mechanical flushing only; does not dissolve capsaicin	Low (oral/dermal); essential (ocular—volume matters more than solvency)	Ocular (primary); dermal (final rinse only)

10. Insurance and Liability

10.1 Product Liability Insurance

Table 10-1. Minimum Insurance Coverage Requirements

COVERAGE TYPE	MINIMUM PER OCCURRENCE	MINIMUM AGGREGATE	NOTES
Commercial General Liability (CGL)	$1,000,000	$2,000,000	Must include products-completed operations coverage
Product Liability (standalone or CGL endorsement)	$1,000,000	$2,000,000	Must cover bodily injury from ingestion, inhalation, and dermal/ocular contact
Workers' Compensation	Statutory limits	Statutory limits	Per state requirements; employer's liability: $500,000/$500,000/$500,000
Umbrella / Excess Liability	$2,000,000	$5,000,000	Recommended for Tier B/C product distribution

10.2 Age Restriction

All Scoville Splice products are restricted to purchasers and consumers aged 18 years and older. Point-of-sale systems must enforce age verification for all direct sales. Online sales require age-gate verification at checkout.

10.3 Liability Waivers

- **Tier A products:** Consumer acknowledgment of warning label is sufficient for retail sale. No additional waiver required.
- **Tier B products:** Signed liability waiver required for all event tastings, demonstrations, and food-service applications. Waiver must include explicit acknowledgment of capsaicin concentration, potential health effects, and assumption of risk.
- **Tier C products:** Signed liability waiver required for all sales, transfers, and demonstrations. Waiver must include acknowledgment of hazmat-equivalent hazard classification, confirmation of training completion, and explicit assumption of risk for all foreseeable exposure scenarios.

10.4 Indemnification

Distributors, retailers, and food-service operators handling Scoville Splice products must maintain their own product liability coverage and execute a mutual indemnification agreement with Scoville Splice prior to receiving product.

11. Training Requirements

Table 11-1. Required Training by Role and Tier

TRAINING MODULE	TARGET AUDIENCE	DURATION	FREQUENCY	TIER REQUIREMENT
Capsaicin Hazard Awareness	All personnel	1 hour	Upon hire + annual refresher	All tiers
PPE Selection and Donning/Doffing	Handlers, processors	2 hours (with practical)	Upon hire + annual refresher	All tiers
Respirator Fit Testing	Handlers (Tier B/C)	30 minutes	Annual (quantitative fit test per 29 CFR 1910.134)	Tier B (N95), Tier C (P100)
First Aid and Decontamination	All personnel	2 hours (with practical)	Upon hire + annual refresher	All tiers
Spill Response and Containment	Handlers, processors, facility staff	1 hour (with tabletop exercise)	Upon hire + annual refresher	Tier B and C
Chemical Handling and Hazmat Awareness	Handlers (Tier C)	4 hours	Upon hire + annual refresher	Tier C
Consumer Interaction and Event Safety	Sales, marketing, event staff	1 hour	Upon hire + per-event briefing	All tiers
Regulatory Compliance Overview	Quality assurance, regulatory affairs	2 hours	Upon hire + when regulations change	All tiers

All training completion must be documented with date, trainer name, trainee signature, and score (where applicable). Training records must be retained for a minimum of 3 years per OSHA recordkeeping requirements (29 CFR 1910.1020).

12. Incident Reporting

12.1 Reporting Requirements

Any exposure incident involving Scoville Splice products must be documented within 24 hours of occurrence. An "incident" is defined as any unplanned exposure resulting in symptoms or any planned exposure resulting in symptoms beyond those expected for the product tier.

12.2 Incident Classification

Table 12-1. Incident Severity Classification

CLASS	DESCRIPTION	EXAMPLES	REPORTING TIMELINE	REPORT TO
I — Minor	Self-limiting symptoms, no medical treatment required	Mild skin irritation, brief eye watering, transient oral discomfort beyond expected	Within 24 hours	Supervisor; incident log
II — Moderate	Symptoms requiring first aid or non-emergency medical treatment	Persistent dermal burns, prolonged eye irritation, nausea/vomiting, moderate respiratory symptoms	Within 8 hours	Supervisor; Safety Officer; incident log
III — Severe	Symptoms requiring emergency medical treatment or hospitalization	Severe bronchospasm, laryngospasm, syncope, anaphylactoid reaction, chemical burns	Immediately (verbal); written within 4 hours	Supervisor; Safety Officer; executive management; OSHA (if recordable per 29 CFR 1904)

12.3 Documentation Requirements

All incident reports must include:

1. Date, time, and location of incident
2. Product involved (Heat number, lot number, form)
3. Personnel involved (names, roles, training status)
4. PPE in use at time of incident
5. Description of events leading to exposure
6. Exposure route, estimated dose, and duration
7. Symptoms observed and their onset timeline
8. First aid and decontamination measures taken

9. Medical treatment provided (if any)
10. Root-cause analysis and corrective actions
11. Photographs of the scene (where applicable)

12.4 Post-Incident Review

All Class II and III incidents trigger a formal post-incident review within 5 business days. The review board consists of the Safety Officer, the supervisor of the affected area, and at least one additional qualified person not directly involved in the incident. Corrective and preventive actions (CAPA) identified in the review must be implemented within 30 days and verified for effectiveness within 90 days.

13. Revision History

VERSION	DATE	AUTHOR	DESCRIPTION OF CHANGES
1.0	October 2026	Scoville Splice / Capsaicin Design Platform	Initial release. Complete SOP covering toxicology, dose-response, three-tier handling system, facility requirements, consumer warnings, US and international regulatory pathways, first aid protocols, insurance and liability, training, and incident reporting.

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